consists of 3 layers, the outer and inner are fleshy (**sarcotesta**), the middle layer is stony called **sclerotesta**. The inner layer remains fused with the nucellus. The nucellus grows out into a beak-like structure and the upper part dissolves and forms a cavity-like structure called **pollen chamber**. A single megaspore mother cell undergoes meiosis to form four haploid megaspores. The lowermost becomes functional and others get degenerated. The nucellus gets reduced in the form of a thin papery layer in mature seeds and encloses the female gametophyte An enlarged megaspore or the embryo-sac is present within the nucellus. An archegonial chamber with 3-6 archegonia are present in the archegonial chamber below the pollen chamber (Figure 2.46).

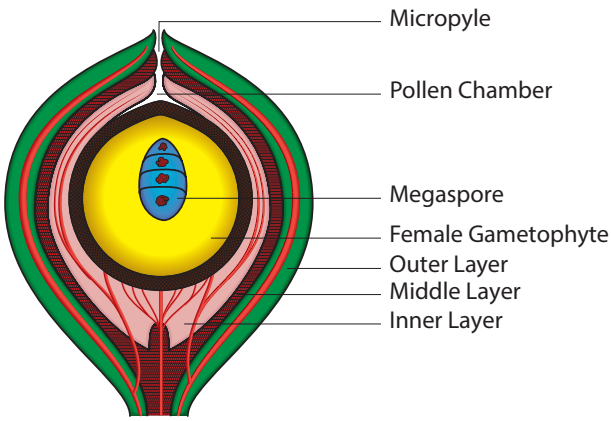



**Figure 2.46:** L.S. of Ovule

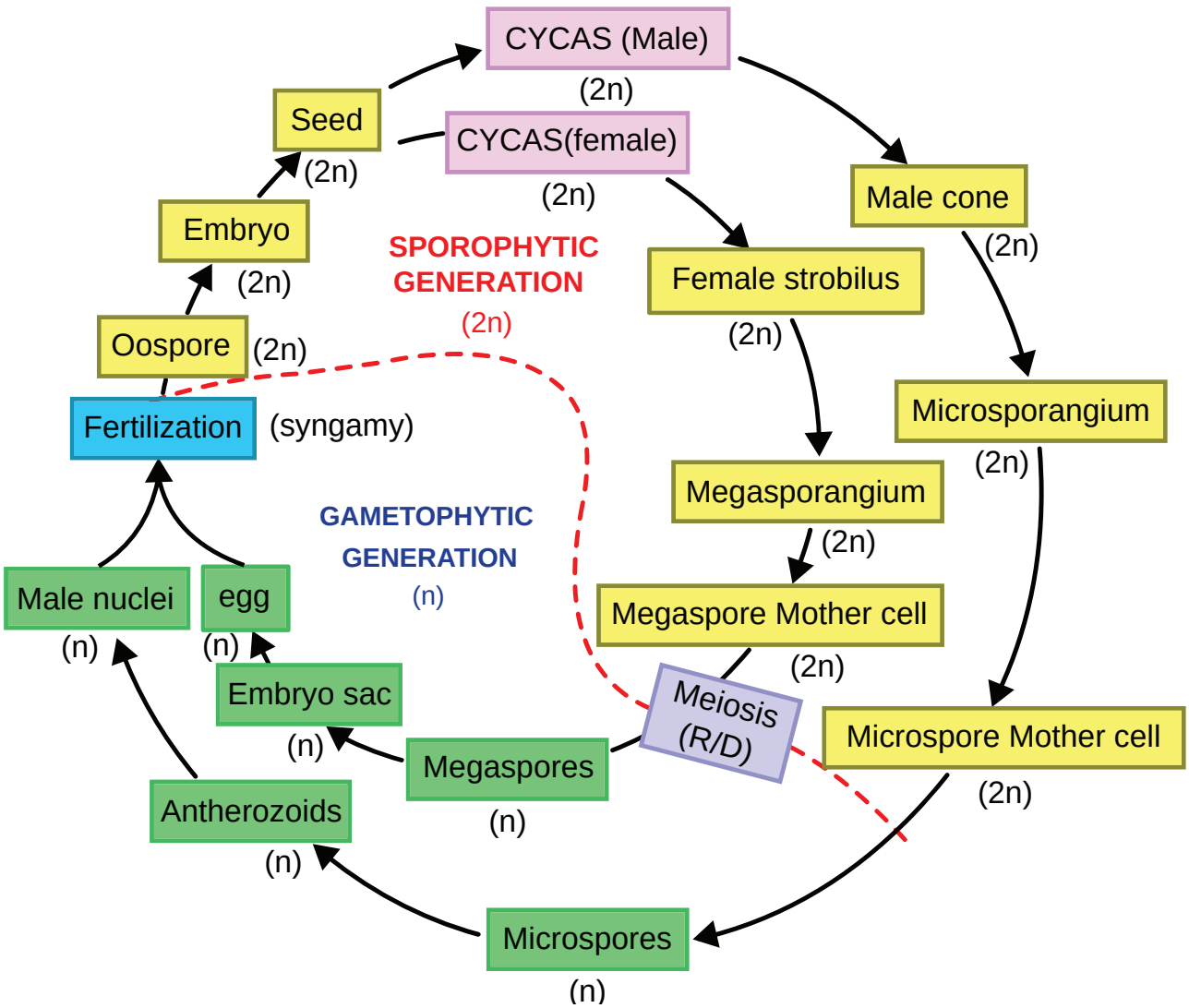



**Figure 2.47:** Life cycle of *Cycas*